PHYSICS-KM-02

Topics Covered: *3 equations of motion (non-gravity + gravity cases)*

Total Count of Questions: *26*

Expected Time of Attempt by Students: *130 min*

1. An object having a velocity 4.0 m/s is accelerated at the rate of 1.2 m/s^2 for 5.0 s. Find the distance travelled during the period of acceleration.

2. A person travelling at 43.2 km/h applies the brake giving a deceleration of 6.0 m/s^2 to his scooter. How far will it travel before stopping?

3. A train starts from rest and moves with a constant acceleration of 2.0 m/s^2 for half a minute. The brakes are then applied and the train comes to rest in one minute. Find
(a) the total distance moved by the train,
(b) the maximum speed attained by the train and
(c) the position(s) of the train at half the maximum speed.

4. A bullet travelling with a velocity of 16 m/s penetrates a tree trunk and comes to rest in 0.4 m. Find the time taken during the retardation.

5. A bullet going with speed 350 m/s enters a concrete wall and penetrates a distance of 5.0 cm before coming to rest. Find the deceleration.

6. A police jeep is chasing a culprit going on a motorbike. The motorbike crosses a turning at a speed of 72 km/h. The jeep follows it at a speed of 90 km/h, crossing the turning ten seconds later than the bike. Assuming that they travel at constant speeds, how far from the turning will the jeep catch up with the bike?

7. A car travelling at 60 km/h overtakes another car travelling at 42 km/h. Assuming each car to be 5.0 m long, find the time taken during the overtake and the total road distance used for the overtake.

8. A ball is projected vertically upward with a speed of 50 m/s. Find (a) the maximum height, (b) the time to reach the maximum height, (c) the speed at half the maximum height. Take g = 10 m/s^2.

9. A ball is dropped from a balloon going up at a speed of 7 m/s. If the balloon was at a height 60 m at the time of dropping the ball, how long will the ball take in reaching the ground?

10. A stone is thrown vertically upward with a speed of 28 m/s.
(a) Find the maximum height reached by the stone.
(b) Find the velocity one second before it reaches the maximum height.
(c) Does the answer of part (b) change if the initial speed is more than 28 m/s such as 40 m/s or 80 m/s?

11. A person sitting on the top of a tall building is dropping balls at regular intervals of one second. Find the positions of the 3rd, 4th and 5th ball when the 6th ball is being dropped.

12. A healthy young man standing at a distance of 7 m from a 11.8 m high building sees a kid slipping from the top floor. With what speed (assumed uniform) should he run to catch the kid at the arms height (1.8 m)?

13. A ball is dropped from a height. If it takes 0.200 s to cross the last 6.00 m before hitting the ground, find the height from which it was dropped. Take g = 10 m/s^2.

14. A ball is dropped from a height of 5 m onto a sandy floor and penetrates the sand up to 10 cm before coming to rest. Find the retardation of the ball in sand assuming it to be uniform.

15. A rock is thrown vertically upwards from ground level at time t = 0. At t = 1.5 seconds, it passes the top of a tall tower and 1 second later it reaches the maximum height. What is the height of the tower?

16. An object falls a distance 'h' from rest. If it travels 0.50 h in last second, find
(a) the time and
(b) the height of the fall. Take t = 10 m/s^2.

17. A car is travelling along a straight road with constant acceleration. At time t = 0, its position x_0 is –4 m and its velocity u is 5 ms^{-1}. After it has moved 50 m, its velocity v is 10 ms^{-1}. Calculate
(i) The acceleration of the car.
(ii) Time taken by the car to cover this 50 m.
(iii) Position of the car after 4s.

18. When an object is projected upwards with some initial velocity from a height h above the ground, it reaches the ground after time t_1. If it is thrown downwards with same velocity, it reaches the ground after time t_2. What time would it take to reach the ground if it is dropped from rest from the same height?

19. An elevator ascends from the ground with uniform speed. At time T_1, a boy drops a marble through the floor. The marble falls with uniform acceleration g = 9.8 m/s^2, and hits the ground T_2 seconds later. If $T_1 = T_2 = 4$ sec, then find the height of the elevator at time T_1.

20. The acceleration due to gravity can be measured by projecting a body upward and measuring the time that it takes to pass two given points in both directions. Show that if the time body takes to pass a horizontal line A in both directions is T_A, and the time to go by a second line B in both the directions is T_B, then assuming that the acceleration is constant, its magnitude is $g = \frac{8h}{T_A^2 - T_B^2}$, where 'h' is the height of the line B above line A.

21. A particle starts from rest with uniform acceleration a. Its velocity after n seconds is v. The displacement of the particle in the last two seconds is :

(a) $\frac{2v(n-1)}{n}$ (b) $\frac{v(n-1)}{n}$

(c) $\frac{v(n+1)}{n}$ (d) $\frac{2v(2n+1)}{n}$

22. A particle has an initial velocity of $(3\hat{i} + 4\hat{j})m/s$ and a constant acceleration of $(4\hat{i} - 3\hat{j})m/s^2$. Its speed after one second will be equal to :-

(a) 0 (b) 10 m/s

(c) $5\sqrt{2}m/s$ (d) 25 m/s

23. A body starts from rest and is uniformly accelerated for 30 s. The distance travelled in the first 10 s is x_1, next 10 s is x_2 and the last 10 s is x_3. Then $x_1 : x_2 : x_3$ is the same as:-

(a) 1 : 2 : 4 (b) 1 : 2 : 5

(c) 1 : 3 : 5 (d) 1 : 3 : 9

24. A particle is projected vertically upwards and it reaches the maximum height H in T seconds. The height of the particle at any time t will be :-

(a) $H - g(t-T)^2$ (b) $g(t-T)^2$

(c) $H - \frac{1}{2}g(t-T)^2$ (d) $\frac{g}{2}(t-T)^2$

25. A particle is moving with uniform acceleration along a straight line . Its velocities at A & B are respectively 7 m/s & 17 m/s. M is mid point of AB. If t_1 is the time taken to go from A to M and t_2 the time taken to go from M to B, the ratio $\frac{t_1}{t_2}$ is equal to:-

(a) 3 : 2 (b) 3 : 1

(c) 2 : 1 (d) 2 : 3

26. A particle moves with uniform acceleration and v_1, v_2 and v_3 denote the average velocities in the three successive intervals of time t_1, t_2 and t_3. Which of the following relations is correct?

(a) $(v_1 - v_2):(v_2 - v_3) = (t_1 - t_2):(t_2 + t_3)$

(b) $(v_1 - v_2):(v_2 - v_3) = (t_1 + t_2):(t_2 + t_3)$

(c) $(v_1 - v_2):(v_2 - v_3) = (t_1 - t_2):(t_1 - t_3)$

(d) $(v_1 - v_2):(v_2 - v_3) = (t_1 - t_2):(t_2 - t_3)$

ANSWERS

1. 35 m

2. 12 m

3. (a) 2.7 km

(b) 60 m/s

(c) 225 m and 2.25 km

4. 0.05 s

5. 12.2×10^5 m/s^2

6. 1.0 km

7. 2s, 38 m

8. (a) 125 m

(b) 5s

(c) 35 m/s

9. 4.3 s

10. (a) 40 m

(b) 9.8 m/s

(c) No

11. 44.1m, 19.6 m and 4.9 m below the top

12. 4.9 m/s

13. 48 m

14. 490 m/s^2

15. 26.25 m

16. $t = 2 + \sqrt{2}\,\text{sec}, h = 58.28m$

17. a = 0.75 m/s^2, t = 6.66 sec, x = 22 meter

18. $\sqrt{t_1 t_2}$

19. 39.2m

20.

21. (a)

22. (c)

23. (c)

24. (c)

25. (a)

26. (b)